

پلتفرم خرید و فروش توکن‌های غیر قابل معاوضه (NFT) مبتنی بر بلاکچین



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1. Introduction

Concept:

This project introduces a decentralized marketplace for Non-Fungible Tokens (NFTs), built upon blockchain technology. The core of the platform is a solid smart contract that autonomously manages all on-chain logic, including asset ownership, sale processes, and financial transactions, ensuring a transparent and immutable environment for trading digital assets.



Problem:

While the NFT market has grown significantly, many existing platforms present considerable challenges for users, such as complex user interfaces and high transaction fees. This creates a barrier for artists, creators, and collectors who are not familiar with blockchain technology, highlighting the need for a more accessible and simplified solution.

Project Goal:

The goal of this project is to develop a seamless and secure platform for trading NFTs that bridges the gap between blockchain's complexity and the user's need for simplicity. We aim to create a decentralized marketplace where the core logic is governed by a transparent smart contract, while providing a user-friendly interface that feels as intuitive as a standard web application.

2. System Design

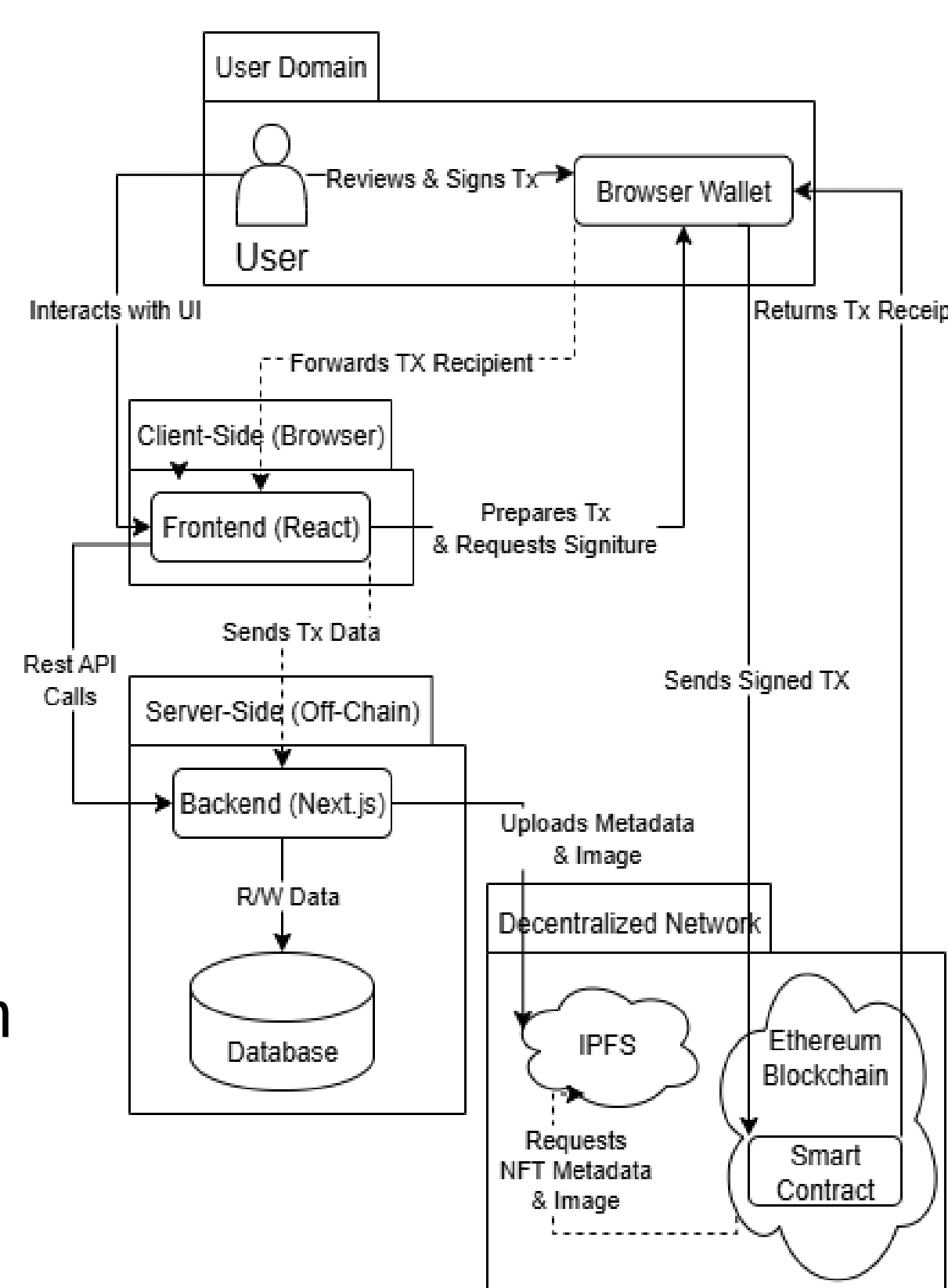
User Requirements:

The system is designed to meet key functional and non-functional requirements for an NFT marketplace. Functionally, it supports wallet-based authentication, the complete token lifecycle (creation, listing, and purchase), and financial operations like fund withdrawals. Non-functional requirements prioritize security through re-entrancy guards and role-based access, and an emergency-stop mechanism in the contract.

Architecture:

The platform uses a multi-layer dApp architecture that separates on-chain and off-chain logic for optimal performance and security.

- On-Chain Layer (Smart Contract):** The core of the system, this contract can be deployed on the Ethereum blockchain. It governs all critical operations, including asset ownership, market rules, and the secure handling of funds.
- Off-Chain Layer (Backend & Frontend):** A Next.js backend manages user data, authentication, and communication with the IPFS storage system. The React-based frontend provides a dynamic user interface that allows users to interact with the backend and, via their wallets, execute transactions on the smart contract.



3. Implementation

Smart Contract:

The core logic is encapsulated in a smart contract developed with Solidity and the Hardhat framework. To ensure security and standardization, it inherits from OpenZeppelin's libraries, implementing the ERC-721 standard for NFTs. Key security patterns like ReentrancyGuard and Pausable are integrated to protect user assets. The contract's public functions (createToken, createMarketSale, withdraw) govern all on-chain market activities, ensuring transparent and autonomous operation.

Backend & Database:

The off-chain backend is built using Next.js API Routes, providing a serverless environment for managing application data. Prisma is utilized as a modern ORM for type-safe interaction with a database. It ensures data integrity through atomic transactions, which are crucial for synchronizing the off-chain database with on-chain events after a transaction is confirmed.

Frontend & Web3 Integration:

The frontend, built with React and Next.js, serves as an intuitive gateway to the blockchain. The connection to the blockchain is handled by a robust Web3 stack:

- Wagmi & Viem:** These libraries provide powerful and efficient hooks for reading from and writing to the smart contract.
- RainbowKit:** Integrated to offer a seamless and user-friendly multi-wallet connection experience.
- NextAuth.js:** Implements secure, passwordless authentication using the SIWE protocol, allowing users to log in simply by signing a message with their wallet.

4. Conclusions

This project successfully achieved its primary goal of designing and implementing a full-stack, decentralized marketplace for NFTs. The final platform provides a user-friendly and secure solution that allows users to manage the entire lifecycle of their digital assets, from creation to sale, within an on-chain environment.

Key achievements include the implementation of a secure Solidity smart contract governing all market logic, an off-chain backend for data management, and a dynamic frontend that ensures a user-friendly experience. By integrating technologies like IPFS for decentralized storage and SIWE for wallet-based authentication, the project delivers a robust and practical example of a modern decentralized application that effectively bridges the gap between blockchain's complexity and user accessibility.

